

CO2-AT1-CSA0911

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

Asif Ali Khan L
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QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

4/5 answered

Q1 Student Registration System 10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2
3 public class StudentRegistrationSystem {
4     public static void main(String[] args) {
5         HashSet<String> students = new HashSet<>();
6         HashMap<String, List<String>> studentCourseMap = new HashMap<>();
7
8         students.add("Asif");
9         students.add("Sakthi");
10        students.add("Ravi");
11
12        studentCourseMap.put("Asif", Arrays.asList("Java", "DBMS"));
13        studentCourseMap.put("Sakthi", Arrays.asList("Python", "OS"));
14        studentCourseMap.put("Ravi", Arrays.asList("C++", "DSA"));
15
16        Iterator<String> studentIterator = students.iterator();
17        while (studentIterator.hasNext()) {
```

Input

stdin input

Output

```
Student: Asif
Courses: Java DBMS
Student: Sakthi
Courses: Python OS
Student: Ravi
Courses: C++ DSA
```

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CO2-AT1-Scenario Based Program**QUESTIONS**

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Q2 E-Commerce Cart System 10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3
4 public class ECommerceCart {
5     public static void main(String[] args) {
6         ArrayList<String> cart = new ArrayList<>();
7
8         cart.add("Laptop");
9         cart.add("Mobile Phone");
10        cart.add("Headphones");
11        cart.add("Keyboard");
12
13        cart.remove("Headphones");
14
15        Iterator<String> it = cart.iterator();
```

Input

stdin input

Output

```
Items in Cart:
Laptop
Mobile Phone
Keyboard
```

CO2-AT1-Scenario Based Program

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10 marks

4/5 answered

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- (a) Name three Map implementations in Java.
(b) Design a schema-like representation using Map for storing ISBN and book details.
(c) Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

Your Code

```
1 import java.util.HashMap;
2 import java.util.Map;
3
4 public class LibraryManagementSystem {
5     public static void main(String[] args) {
6         HashMap<String, String> library = new HashMap<>();
7
8         library.put("978-0134685991", "Effective Java by Joshua Bloch");
9         library.put("978-0596009205", "Head First Design Patterns by Eric Freeman");
10        library.put("978-0132350884", "Clean Code by Robert C. Martin");
11
12        String isbnToSearch = "978-0596009205";
13        if (library.containsKey(isbnToSearch)) {
14            System.out.println("Book found: " + library.get(isbnToSearch));
15        }
16    }
17 }
```

Input

stdin input

Output

```
Book found: Head First Design
Patterns by Eric Freeman
All Books in Library:
ISBN: 978-0596009205 -> Head
First Design Patterns by Eric
```

CO2-AT1-Scenario Based Program

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10 marks

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10 marks

4/5 answered

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- (a) List three advantages of Generics in Collections.
(b) Represent how employee data is stored using List with Generics.
(c) Write a Java program using Iterator to:

- Store employee objects
- Display employees with salary > 50,000

Your Code

```
1 import java.util.*;
2
3 class Employee {
4     private String name;
5     private double salary;
6
7     public Employee(String name, double salary) {
8         this.name = name;
9         this.salary = salary;
10    }
11
12    public String getName() {
13        return name;
14    }
15
16    public double getSalary() {
```

Input

stdin input

Output

```
Name: John, Salary: 60000.0
Name: Doe, Salary: 75000.0
Name: Alex, Salary: 55000.0
```

CO2-AT1-Scenario Based Program

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5/5 answered

Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- (a) Identify three suitable Collection types for this system.
(b) Design a structure using Set and Map for voters and vote counting.
(c) Write a Java program using HashSet and HashMap to:
- Record votes
 - Prevent duplicate voting
 - Display results

Your Code

```
1 import java.util.*;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner scanner = new Scanner(System.in);  
6         Set<String> voters = new HashSet<>();  
7         Map<String, Integer> voteCounts = new HashMap<>();  
8  
9         while (scanner.hasNextLine()) {  
10             String input = scanner.nextLine().trim();  
11             if (input.equalsIgnoreCase("exit") || input.isEmpty()) {  
12                 break;  
13             }  
14  
15             String candidate = input;  
16
```

Input

Alice
Bob
Charlie
exit

Output

```
Vote recorded for: Alice  
Vote recorded for: Bob  
Vote recorded for: Charlie  
--- Voting Results ---  
Alice: 1
```